

/* RNEP BSC-IV SEM

Title: Practical on Differential Equations and Integral Transforms

Course Code: C4MAT1P1

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/*LAB-1 Program to verify the exactness of a differential equation*/

M:x.y^2\$

N:x^2.y\$

eq:M.'del(x)+N.'del(y)=0;

My=My:diff(M,y);

Nx=Nx:diff(N,x);

if My=Nx then

print("the given D.E is exact")

else

print("the given D.E is not exact")\$

$$x^2 y \text{ del}(y) + x y^2 \text{ del}(x) = 0 \quad \quad \quad My = 2xy \quad \quad \quad Nx = 2xy$$

y

the given D.E is exact

/*LAB-2 Program to get the solutions of differential equations that are solvable for x*/

kill(all)\$

depends([x,p],y)\$

eq:x-y/(3.p)+2.p.y^2=0;

eq1:diff(eq,y);

eq2:subst(diff(x,y)=1/p,eq1);

eq3:solve(eq2,diff(p,y));

s:ode2(eq3[1],p,y);

disp("The solution of d.e. is")\$

sol:subst(s,eq);

$$2py^2 - \frac{y}{3p} + x = 0 \quad \quad \quad 2\left(\frac{d}{dy}p\right)y^2 + \frac{\left(\frac{d}{dy}p\right)y}{3p^2} + 4p$$

$$y + \frac{d}{dy}x - \frac{1}{3p} = 0 \quad \quad \quad 2\left(\frac{d}{dy}p\right)y^2 + \frac{\left(\frac{d}{dy}p\right)y}{3p^2} + 4py + \frac{2}{3p}$$

$$= 0 \quad \left[\frac{d}{dy}p = -\left(\frac{2p}{y}\right) \right] \quad \quad \quad p = \frac{\%c}{y^2}$$

The solution of d.e. is
$$-\left(\frac{y^3}{3\%c}\right) + x + 2\%c = 0$$

/*LAB-2 Program to get the solutions of differential equations that are solvable for y*/

```
kill(all)$
depends([y,p],x)$
eq:y+p*x-x^4*p^2=0;
eq1:diff(eq,x);
eq2:subst(diff(y,x)=p,eq1);
eq3:solve(eq2,diff(p,x));
s:radcan(ode2(eq3[1],p,x));
disp("The solution of d.e. is")$
sol:subst(s,eq);
```

$$y - p^2 x^4 + p x = 0 \quad \frac{d}{dx} y - 2p \left(\frac{d}{dx} p \right) x^4 - 4p^2 x^3 + \left(\frac{d}{dx} p \right) x + p = 0$$

$$-\left(2p \left(\frac{d}{dx} p \right) x^4 \right) - 4p^2 x^3 + \left(\frac{d}{dx} p \right) x + 2$$

$$p = 0 \quad \left[\frac{d}{dx} p = - \left(\frac{2p}{x} \right) \right] \quad p = \frac{\%c}{x^2}$$

The solution of d.e. is
$$y + \frac{\%c}{x} - \%c^2 = 0$$

/*LAB-2 Program to get the solutions of differential equations that are solvable for p*/

```
kill(all)$
a:solve(p^2-5*p-6,p);
n:length(a);
for i:1 thru n do(
b[i]:lhs(a[i])-rhs(a[i]),
eq[i]:subst(p='diff(y,x),b[i]),
s[i]:ode2(eq[i],y,x),
disp(s[i]))$
disp("The solution of d.e. is")$
prod:1$
for i:1 thru n do(
prod:prod*lhs(s[i]-rhs(s[i]))$
disp(prod=0)$ ;;;
```

$$[p = 6, p = -1] \quad 2$$

$$y = 6x + \%c$$

$$y = \%c - x$$

The solution of d.e. is

$$(y - 6x - \%c)(y + x - \%c) = 0$$

/*LAB-3 Program to find the Complementary Function and Particular Integral of linear differential equations with constant coefficients*/

```
kill(all)$
eq:'diff(y,x,2)+5*diff(y,x)+6*y=3*%e^(2*x);
sol:ode2(eq,y,x)$
cf:rhs(ode2(lhs(eq),y,x))$
print("CF =",cf)$
pi:rhs(sol-cf)$
print("PI =",pi)$
print("The solution is y=CF+PI")$
print(sol)$
```

$$\frac{d^2}{dx^2} y + 5 \left(\frac{d}{dx} y \right) + 6 y = 3 e^{2x}$$

$$CF = \%k1 \%e^{-(2x)} + \%k2 \%e^{-(3x)}$$

$$PI = \frac{3 e^{2x}}{20}$$

The solution is y=CF+PI

$$y = \frac{3 e^{2x}}{20} + \%k1 \%e^{-(2x)} + \%k2 \%e^{-(3x)}$$

/*LAB-4 Program to get solutions of linear partial differential equations of type1 and type4

/*P.D.E of form f(p,q)=0*/

```
kill(all)$
eq:p^2+q^2=4;
s:z=a*x+b*y+c$
h:subst([p=a,q=b],eq);
k:solve(h,b);
print("the required complete solution is")$
subst(k,s);
/*P.D.E of form z=px+qy+f(p,q) */
eq:z=p*x+q*y+p^2+q^2;
print("the required complete solution is")$
print(p=a,q=b,eq);
```

$$\begin{aligned} q^2 + p^2 &= 4 & b^2 + a^2 &= 4 \\ \left[b = -\sqrt{4-a^2}, b = \sqrt{4-a^2} \right] \end{aligned}$$

the required complete solution is $z = -\left(\sqrt{4-a^2} y\right) + a x + c$

$$z = q y + p x + q^2 + p^2$$

the required complete solution is

$$p = a \quad q = b \quad z = qy + px + \frac{q^2}{2} + \frac{p^2}{2} \quad z = qy + px + \frac{q^2}{2} + \frac{p^2}{2}$$

/*LAB-4 Program to get solutions of linear partial differential equations of type3*/

/*pde of the form f(p,x)=g(q,y)*/

/*eq:(p+x)/p=-y/q;*/

assume (a>0)\$

eq:p^2-x^2=y^2-q^2;

p1:lhs(eq)=a;

q1:rhs(eq)=a;

h:solve(p1,p);

k:solve(q1,q);

m:length(h)\$

n:length(k)\$

print("the required complete solution is")\$

z=integrate(rhs(h[m]),x)+integrate(rhs(k[n]),y)+b;

$$\begin{aligned} p^2 - x^2 = y^2 - q^2 & \quad p^2 - x^2 = a & \quad y^2 - q^2 = a \\ \left[p = -\sqrt{x^2 + a}, p = \sqrt{x^2 + a} \right] & \quad \left[q = -\sqrt{y^2 - a}, q = \sqrt{y^2 - a} \right] \end{aligned}$$

the required complete solution is $z = -$

$$\left(\frac{a \log \left(2 \sqrt{y^2 - a} + 2y \right)}{2} \right) + \frac{y \sqrt{y^2 - a}}{2} + \frac{a \operatorname{asinh} \left(\frac{x}{\sqrt{a}} \right)}{2} + \frac{x \sqrt{x^2 + a}}{2} + b$$

/*LAB-4 Program to get solutions of linear partial differential equations of type 2*/

/*P.D.E of form f(p,q,z)=0*/

kill(all)\$

eq:p^3+q^3=3.p.q.z;

eq1:subst([p='diff(z,u),q=a.'diff(z,u)],eq);

k:solve(eq1,'diff(z,u));

n:length(k);

print("the required complete solution is")\$

s:ode2(k[n-1],z,u)\$

subst(u=x+a.y,s);

$$\begin{aligned} q^3 + p^3 = 3 p q z & \quad a^3 \left(\frac{d}{d u} z \right)^3 + \left(\frac{d}{d u} z \right)^3 = 3 a z \\ \left(\frac{d}{d u} z \right)^2 & \quad \left[\frac{d}{d u} z = \frac{3 a z}{a^3 + 1}, \frac{d}{d u} z = 0 \right] \end{aligned}$$

$$z = \%c \%e^{\frac{3 a (a y + x)}{a^3 + 1}}$$

the required complete solution is $z = \%c \%e^{\frac{3 a (a y + x)}{a^3 + 1}}$

/*LAB-4 Program to get solutions of linear partial differential equations of type $Pp+Qq=R$ Lagrange's method - Grouping Method*/
 /*pde of the form $Pp+Qq=R$ using grouping method*/

```
kill(all)$
P:y.z$
Q:z.x$
R:x.y$
eq:P.p+Q.q=R;
print("the auxillary equation are")$
print(dx/P,"=",dy/Q,"=",dz/R)$
eq1:1/P=1/Q.'diff(y,x);    /*taking 1 and 2 terms*/
ode2(eq1,y,x);
s1:solve(%,%c)$
eq2:1/P=1/R.'diff(z,x);    /*taking 1 and 3 terms*/
ode2(eq2,z,x);
s2:solve(%,%c)$
print(" the required solution is ")$
f(rhs(s1[1]),rhs(s2[1]))=0;
      p y z + q x z = x y
the auxillary equation are
```

$$\frac{dx}{yz} = \frac{dy}{xz} = \frac{dz}{xy}$$

$$\frac{1}{yz} = \frac{\frac{d}{dx}y}{xz} \quad \frac{y^2}{2} = \frac{x^2}{2} + \%c$$

$$\frac{1}{yz} = \frac{\frac{d}{dx}z}{xy} \quad \frac{z^2}{2} = \frac{x^2}{2} + \%c$$

the required solution is

$$f\left(\frac{y^2 - x^2}{2}, \frac{z^2 - x^2}{2}\right) = 0$$

/*LAB-4 Program to get solutions of linear partial differential equations of type $Pp+Qq=R$ Lagrange's method - Multipliers method*/
 /*pde of the form $Pp+Qq=R$ using multipliers method*/

```
kill(all)$
P:x*(y-z)$
Q:y*(z-x)$
R:z*(x-y)$
eq:P*p+Q*q=R;
print("the auxillary equations are")$
print(dx/P,"=",dy/Q,"=",dz/R)$
/*choose l,m,n as multiplier s.t LP+mQ+nR=0*/
l:1$
m:1$
n:1$
if ratsimp(l*P+m*Q+n*R)=0 then
s1:integrate(l,x)+integrate(m,y)+integrate(n,z)=a;
/*chose L,M,N as multipliers s.t LP+MQ+NR=0*/
L:1/x$
M:1/y$
N:1/z$
if ratsimp(L*P+M*Q+N*R)=0 then
s2:integrate(L,x)+integrate(M,y)+integrate(N,z)=b;
print("the required solution is")$
f(lhs(s1),lhs(s2))=0;
```

$$q y (z-x) + p x (y-z) = (x-y) z$$

the auxillary equations are

$$\frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{(x-y)z} \quad z+y+x=a \quad \log(z)$$

$$+ \log(y) + \log(x) = b$$

the required solution is $f(z+y+x, \log(z) + \log(y) + \log(x)) = 0$

```

/* LAB-5 Program to find the solution of total differential equation Pdx+Qdy+Rdz=0*/
/*Solve total D.E (2x^2+2xy+2xz^2+1)dx+dy+2z dz=0*/
kill(all)$
P:2*x^2+2*x*y+2*x*z^2+1$
Q:1$ R:2*z$
eqn1:P*'del(x)+Q*'del(y)+R*'del(z);
l:P*(diff(Q,z)-diff(R,y))+Q*(diff(R,x)-diff(P,z))+R*(diff(P,y)-diff(Q,x))$ if l#0 then
disp("The given equation is not integrable, hence there is no solution")
else
disp ("The given equation is integrable")$
print("x as constant implies dx=0 and the given d.e becomes")$
eq1:Q+R*'diff(z,y)=0; print("solution of the above is")$
s1:ode2(eq1,z,y)$ s2:subst([%c=f(x)],s1)$
a1:rhs(s2)-lhs(s2)=0; s3:lhs(a1)-f(x)$
print("differentiating we get ")$ eq2:diff(a1); eq3:lhs(eq2)-rhs(eq2)$
P1:coeff(lhs(eq3),del(x))$
Q1:coeff(lhs(eq3),del(y))$
R1:coeff(lhs(eq3),del(z))$
print("comparing with the given equation and we get ")$
a2:P-P1=0;
eq:substpart(factor(piece),lhs(a2),[1,2])=0;
eq4:radcan(subst(s3=-f(x),eq=0));
print("the required solution is")$
s4:radcan(ode2(eq4,f(x),x));
soln:subst(f(x)=-s3,s4);

```

$$\text{del}(y) + 2z \text{del}(z) + (2xz^2 + 2xy + 2x^2 + 1) \text{del}(x)$$

The given equation is integrable

x as constant implies dx=0 and the given d.e becomes $2z$

$$\left(\frac{d}{dy} z \right) + 1 = 0$$

solution of the above is $z^2 + y + f(x) = 0$

differentiating we get $2z \text{del}(z) + \text{del}(y) + \left(\frac{d}{dx} f(x) \right) \text{del}(x) = 0$

comparing with the given equation and we get $2xz^2 + 2xy$

$$-\frac{d}{dx} f(x) + 2x^2 + 1 = 0 \quad 2x(z^2 + y) - \frac{d}{dx} f(x) + 2x^2 + 1 = 0$$

$$-\left(\frac{d}{dx} f(x) \right) - 2xf(x) + 2x^2 + 1 = 0 = 0$$

the required solution is $f(x) = e^{-x^2} \left(\frac{x^2}{2} + \frac{1}{2} \right) -$


```

/*LAB-6 pde of the form f(p,q,x,y,z)=0 using charpits method*/
kill(all)$
eq:(p^2+q^2).y-q.z;
fx:diff(eq,x);
fy:diff(eq,y);
fz:diff(eq,z);
fp:diff(eq,p);
fq:diff(eq,q);
print("the auxillary equation are")$
print(dp/(fx+p.fz),"=",dq/(fy+q.fz),"=",dx/-fp,"=",dy/-fq,"=",dz/(-p.fp-q.fq))$
eq1:'diff(p,q)=(fx+p.fz)/(fy+q.fz); /*taking 1 and 2 terms*/
ode2(eq1,p,q);
s:subst(%c=-a/2,%);
av:solve(s,a);
s1:subst(q^2+p^2=lhs(av[1]),eq);
s2:solve(s1,q);
s3:subst(q=rhs(s2[1]),av);
pv:solve(s3,p);
eq2:dz=rhs(pv[2]).dx+rhs(s2[1]).dy;
eq3:z.factor(eq2);
eq4:(eq3-a.y.dy)/coeff(rhs(eq3),dx);
print(" the required solution")$
s4:subst([z^2-a.y^2=u,z.dz-a.y.dy=du],eq4);
sol:integrate(lhs(coeff(s4,du)),u)=integrate(rhs(coeff(s4,dx)),x)+b;
subst(u=z^2-a.y^2,sol);

```

$$(q^2 + p^2)y - qz = 0 \quad q^2 + p^2 = -q \quad 2$$

$$p y \quad 2 q y - z$$

the auxillary equation are

$$-\left(\frac{dp}{p q}\right) = \frac{dq}{p^2} = -\left(\frac{dx}{2 p y}\right) = \frac{dy}{z - 2 q y} =$$

$$\frac{dz}{-(q(2 q y - z)) - 2 p^2 y} \quad \frac{d}{d q} p = -\left(\frac{q}{p}\right) \quad -\left(\frac{p^2}{2}\right) =$$

$$\frac{q^2}{2} + \%c \quad -\left(\frac{p^2}{2}\right) = \frac{q^2}{2} - \frac{a}{2} \quad \left[a = q^2 + p^2\right] \quad a y$$

$$-q z \quad \left[q = \frac{a y}{z}\right] \quad \left[a = \frac{a^2 y^2}{z^2} + p^2\right]$$

$$\left[p = -\left(\frac{\sqrt{a z^2 - a^2 y^2}}{z}\right), p = \frac{\sqrt{a z^2 - a^2 y^2}}{z}\right] \quad dz =$$

/* LAB-7 Program to find the Laplace transforms of some standard functions*/

kill(all)\$

f(t):=cos(5*t);

print("L[f(t)=",laplace(f(t),t,s))\$

f(t):=cos(5 t)

$$L[f(t)] = \frac{s}{s^2 + 25}$$

/* LAB-7 Program to find the Laplace transforms of some periodic functions*/

kill(all)\$

f(t):=t;

T:2\$

k:integrate(%e^(-s*t)*f(t),t,0,T)\$

print("Laplace transform of periodic function with period T=",T," is ")\$

print("L[f(t)=",1/(1-%e^(-s*T))*k)\$

f(t):=t

Laplace transform of periodic function with period T= 2 is

$$L[f(t)] = \frac{\frac{1}{s} - \frac{(2s+1)e^{-(2s)}}{s^2}}{1 - e^{-(2s)}}$$

/* LAB-8 Program to verify Convolution theorem*/

assume(t>0)\$

f(t):=1;

g(t):=sin(t);

k1:laplace(f(t),t,S)\$

k2:laplace(g(t),t,S)\$

RHS=rhs:factor(k1*k2);

k3:integrate(f(t-u)*g(u),u,0,t)\$

k4:laplace(k3,t,S)\$

LHS=lhs:factor(k4);

if lhs=rhs then

print("the convolution therom is verified")

else

print("the convolution therom is not verified")\$

$$f(t) := 1 \quad g(t) := \sin(t) \quad \text{RHS} = \frac{1}{S(S^2 + 1)}$$

$$\text{LHS} = \frac{1}{s(s^2 + 1)}$$

the convolution theorem is verified

/* LAB-9 Program to solve ordinary linear differential equations using Laplace transforms*/

```
kill(all)$
eq:'diff(y,t,2)+9*y=sin(t);
y1(k):=0$
yd(k):=0$
d1:s*L(y(t))-y1(0)$
d2:s^2*L(y(t))-s*y1(0)-yd(0)$
eq1:d2+9*L(y(t))=laplace(sin(t),t,s);
h:solve(eq1,L(y(t)));
L(y(t))=h1:partfrac(rhs(h[1]),s);
print("the solution of differential equation is y=",ilt(h1,s,t))$
```

$$\frac{d^2}{dt^2} y + 9y = \sin(t) \qquad s^2 L(y(t)) + 9L(y(t)) = \frac{1}{s^2 + 1}$$

$$\left[L(y(t)) = \frac{1}{s^4 + 10s^2 + 9} \right] \qquad L(y(t)) = \frac{1}{8(s^2 + 1)} -$$

$$\frac{1}{8(s^2 + 9)}$$

the solution of differential equation is $y = \frac{\sin(t)}{8} - \frac{\sin(3t)}{24}$

/* LAB-9 Program to solve ordinary linear differential equations using Laplace transforms*/

```
kill(all)$
eq:'diff(y,t,2)+4*diff(y,t)+4*y=exp(-t);
y1(k):=0$
yd(k):=0$
d1:s*L(y(t))-y1(0)$
d2:s^2*L(y(t))-s*y1(0)-yd(0)$
eq1:d2+4*d1+4*L(y(t))=laplace(exp(-t),t,s);
h:solve(eq1,L(y(t)));
L(y(t))=factor(rhs(h[1]));
L(y(t))=h1:partfrac(rhs(h[1]),s);
print("the solution of differential equation is y=",ilt(h1,s,t))$
```

$$\frac{d^2}{dt^2} y + 4 \left(\frac{d}{dt} y \right) + 4y = e^{-t} \qquad s^2 L(y(t)) + 4s L(y(t))$$

$$+ 4 L(y(t)) = \frac{1}{s+1} \quad \left[L(y(t)) = \frac{1}{s^3 + 5s^2 + 8s + 4} \right] \quad L(y(t))$$

$$= \frac{1}{(s+1)(s+2)^2} \quad L(y(t)) = -\left(\frac{1}{s+2}\right) - \frac{1}{(s+2)^2} + \frac{1}{s+1}$$

the solution of differential equation is $y = e^{-t} - t e^{-(2t)} - \frac{1}{2} e^{-(2t)}$

/* LAB-10 Program to find full range Fourier series of some simple functions with period 2π

/*Fourier series expansion in interval (-π,π) with period 2π*/

kill(all)\$

declare(n,integer)\$

%piargs:true\$

/*f(x):=read("enter the f(x)=")*/

f(x):=x^2;

l:%pi\$

print("the fourier coefficient are")\$

a_0=a0:integrate(f(x),x,l,-l)/l;

a_n=an:integrate(f(x)*cos(n*%pi*x/l),x,-l,l)/l;

b_n=bn:integrate(f(x)*sin(n*%pi*x/l),x,l,l)/l;

print("the fourier series is")\$

f(x)=a0/2+sum(an*cos(n*%pi*x/l),n,1,inf)+sum(bn*sin(n*%pi*x/l),n,1,inf);

$$f(x) = x^2$$

the fourier coefficient are

$$a_0 = -\left(\frac{2\pi^2}{3}\right) \quad a_n = \frac{4(-1)^n}{n^2}$$

$$b_n = 0$$

the fourier series is

$$x^2 = 4 \left(\sum_{n=1}^{\infty} \left(\frac{(-1)^n \cos(nx)}{n^2} \right) \right) - \frac{\pi^2}{3}$$

```
/* LAB-10 Program to find full range Fourier series of some simple functions with period 2L
/*Fourier series expansion in interval (-l,l) with period 2l*/
```

```
kill(all)$
declare(n,integer)$
%piargs:true$
f(x):=x;
l:1$
print("the fourier coefficient are")$
a_0=a0:integrate(f(x),x,-l,l)/l;
a_n=an:integrate(f(x)*cos(n*%pi*x/l),x,-l,l)/l;
b_n=bn:ratsimp(integrate(f(x)*sin(n*%pi*x/l),x,-l,l)/l);
print("the fourier series is")$
f(x)=a0/2+sum(an*cos(n*%pi*x/l),n,1,inf)+sum(bn*sin(n*%pi*x/l),n,1,inf);
f(x):=x
```

the fourier coefficient are $a_0 = 0$ $a_n = 0$ $b_n = -$

$$\left(\frac{2(-1)^n}{\pi n} \right)$$

the fourier series is
$$x = - \left(\frac{2 \sum_{n=1}^{\infty} \left(\frac{(-1)^n \sin(\pi n x)}{n} \right)}{\pi} \right)$$

```
/* LAB-10 Fourier series expansion in interval (-π,π) with period 2π or 2L */
```

```
kill(all)$
load('fourie')$
f:x;
l:1$ /* l:%pi for period 2π */
f=totalfourier(f,x,l);
```

x $a_0 = 0$ $a_n = 0$ $b_n = 2$

$$\left(\frac{\sin(\pi n)}{\pi^2 n^2} - \frac{\cos(\pi n)}{\pi n} \right)$$

$a_0 = 0$ $a_n = 0$ $b_n = -$

$$\left(\frac{2(-1)^n}{\pi n} \right)$$

$$x = - \left(\frac{2 \sum_{n=1}^{\infty} \left(\frac{(-1)^n \sin(\pi n x)}{n} \right)}{\pi} \right)$$

/* LAB-11 Program to find half range sine series of some simple functions*/

load('fourie')\$

f(x):=%pi-x;

l:%pi\$

fs:foursin(f(x),x,l);

foursimp(fs);

f(x)=fourexpend(fs,x,l,inf);

$$f(x) := \pi - x \quad b_n = \frac{2 \left(\frac{\pi}{n} - \frac{\sin(\pi n)}{n^2} \right)}{\pi} \quad [\%t13]$$

$$b_n = \frac{2}{n} \quad [\%t14] \quad \pi - x = 2 \sum_{n=1}^{\infty} \left(\frac{\sin(n x)}{n} \right)$$

/* LAB-11 Program to find half range cosine series of some simple functions*/

load('fourie')\$

f(x):=%pi-x;

l:%pi\$

fc:fourcos(f(x),x,l);

fc1:foursimp(fc);

f(x)=fourexpend(fc,x,l,inf);

$$f(x) := \pi - x \quad a_0 = \frac{\pi}{2} \quad a_n = \frac{2 \left(\frac{1}{n^2} - \frac{\cos(\pi n)}{n^2} \right)}{\pi}$$

$$[\%t19, \%t20] \quad a_0 = \frac{\pi}{2} \quad a_n = - \left(\frac{2 \left((-1)^n - 1 \right)}{\pi n^2} \right)$$

$$[\%t21, \%t22] \quad \pi - x =$$

$$\frac{2 \sum_{n=1}^{\infty} \left(\left(\frac{1}{n^2} - \frac{(-1)^n}{n^2} \right) \cos(n x) \right)}{\pi} + \frac{\pi}{2}$$

/* LAB-11 Program to find half range cosine series of some simple functions*/

```
kill(all)$
declare(n,integer)$
%piargs:true$
f(x):=%pi-x;
l:%pi$
print("the fourier coefficent are")$
a_0=a0:2:integrate(f(x),x,0,l)/l;
a_n=an:2:integrate(f(x)*cos(n*%pi*x/l),x,0,l)/l;
print("the half range cosine series is")$
f(x)=a0/2+sum(an*cos(n*%pi*x/l),n,1,inf);
```

$$f(x) := \pi - x$$

the fourier coefficent are $a_0 = \pi$ $a_n =$

$$\frac{2 \left(\frac{1}{n^2} - \frac{(-1)^n}{n^2} \right)}{\pi}$$

the half range cosine series is $\pi - x =$

$$\frac{2 \sum_{n=1}^{\infty} \left(\left(\frac{1}{n^2} - \frac{(-1)^n}{n^2} \right) \cos(n x) \right)}{\pi} + \frac{\pi}{2}$$

/* LAB-11 Program to find half range sine series of some simple functions*/

```
kill(all)$
declare(n,integer)$
%piargs:true$
f(x):=x;
l:1$
print("the fourier coefficent is")$
b_n=bn:2:ratsimp(integrate(f(x)*sin(n*%pi*x/l),x,0,l)/l);
print("the half range sine series is")$
f(x)=sum(bn*sin(n*%pi*x/l),n,1,inf);
```

$$f(x) := x$$

the fourier coefficent is $b_n = - \left(\frac{2 (-1)^n}{\pi n} \right)$

the half range sine series is

$$x = - \left(\frac{2 \sum_{n=1}^{\infty} \left(\frac{(-1)^n \sin(\pi n x)}{n} \right)}{\pi} \right)$$

/* LAB-12 Program to find sine Fourier transforms*/

kill(all)\$

declare(n,integer)\$

%piargs:true\$

f(x):=x;

l:1;

F_S(n)=(integrate(f(x)·sin(n·%pi·x),x,0,l));

$$f(x) := x \quad 1 \quad F_S(n) = - \left(\frac{(-1)^n}{\pi n} \right)$$

/* LAB-12 Program to find cosine Fourier transforms*/

kill(all)\$

declare(n,integer)\$

%piargs:true\$

f(x):=x;

l:1;

F_C(0)=integrate(f(x),x,0,l);

F_C(n)=integrate(f(x)·cos(n·%pi·x),x,0,l);

$$f(x) := x \quad 1 \quad F_C(0) = \frac{1}{2} \quad F_C(n) = \frac{(-1)^n}{\pi^2 n^2} -$$

$$\frac{1}{\pi^2 n^2}$$